

East Beijing Road

Haoran Chang
York University
Canada
hchang68@yorku.ca



Abstract

East Beijing Road is an asymmetric VR installation that reimagines the stories of a building with a history spanning over 100 years in Shanghai, China, whose residents were compelled to move out due to an ongoing gentrification policy. The project addresses the perceptions of the past and present, the fluidity of archives, and the meanings of ruins. A synthetic space was reconstructed based on 3D photogrammetry models captured in this building. The audience, both inside and outside VR, must use distinct interfaces and experience different visual presentations to collaboratively unfold the story. Old and new, digital and analog, real and fictional, and virtual and physical elements interweave. The asymmetrical design in interactive storytelling blends different times and spaces, allowing the re-enactment of the past through audience collaborative

actions, raising questions about the relationship between interactivity and the visibility of the past. Asymmetry VR can be defined as “co-located users access the same virtual environment using different kinds of technology” [Ouverson and Gilbert 2021]. This project deploys the characteristic of asymmetric VR and provides an experience of unstable, unbalanced, and incomplete information among audiences inside and outside VR. The imbalances and incompleteness in media design are a metaphor for the fragmented time and space inherent in the archival practice of unveiling the past in the digital epoch. No complete information is visible without collaboration between the audiences inside and outside VR. It is an embodied experience that the past can only be revealed through collaborative interaction. The story of the building on East Beijing Road becomes a form of repertoire because of the collaborative engagements of the audiences.

Keywords

Virtual Reality, Asymmetric VR, Interactive Storytelling, Archive

ACM Reference Format:

Haoran Chang. 2024. East Beijing Road. In *SIGGRAPH Asia 2024 XR (SA XR '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681759.3688908>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SA XR '24, December 03–06, 2024, Tokyo, Japan
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1141-1/24/12
<https://doi.org/10.1145/3681759.3688908>

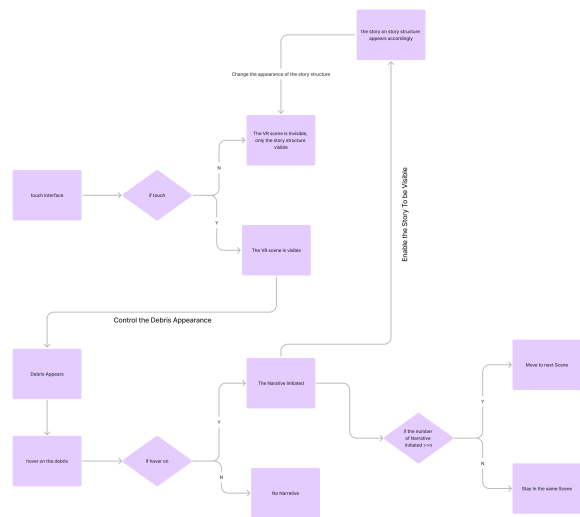


Figure 1: Asymmetric VR System Flowchart

1 Asymmetric Design

In this asymmetric installation, the audience inside and outside the VR environment must work together to reveal the stories in different scenes. The player outside the VR can touch a physical interface to make “debris” appear inside the VR. When the “debris” is enabled, the audience inside the VR can use a controller to hover over it, triggering the narrative accordingly. The audience inside the VR sees a colorfully rendered environment based on the original building on East Beijing Road in Shanghai, China. When the audience outside the VR does not touch the physical sensor, they see film footage rendered from the VR scene. However, when the audience outside the VR touches the physical sensor, they briefly see what the audience inside the VR sees. This asymmetric design, with its interaction between inside and outside the VR, reflects the fragmented nature of archives, questions the relationship between visible and invisible pasts, reimagines ruins in different modalities, and reconsiders the role of technology in re-enactment.

2 Discussion

Asymmetric VR provides a unique mechanism to capture the dynamism and performativity of the archive. Compared to symmetric media design, the power dynamics in asymmetric media facilitate collaborative engagement, offering an opportunity in interactive storytelling to connect distinctive elements, such as disparate times and spaces. In *East Beijing Road*, the asymmetric design reflects the theme that the displacement caused by ongoing gentrification in the metropolitan city transforms the city into an urban palimpsest, layered with heterogeneous times and spaces. This mechanism also has potential in the interactive storytelling of other contexts, portraying contrasts such as old versus new, visible versus invisible, and past versus present, encompassing multiple locations, times, and protagonists.



Figure 2: Reconstructed 3D Environment Based on the Photogrammetry Data Captured Inside the Building

References

Kaitlyn M. Ouverson and Stephen B. Gilbert. 2021. A Composite Framework of Co-located Asymmetric Virtual Reality. 5, CSCW1 (2021). <https://doi.org/10.1145/3449079>